

Probability & Statistics S2 - Resit

Total: 20 questions (20 points).

Duration: 1 HOUR

Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

Instructions: Choose **ONE** correct answer for each question.

- 1 pt for each good answer
- -0.25 pt for a wrong answer
- 0 pt for no answer

Q1. Let A, B be events with $P(A)=0.6$, $P(B)=0.5$, $P(A \cap B)=0.4$. What is $P(A \cup B)$?

- A. 0.7
- B. 0.9
- C. 1.1
- D. 0.5

Q2. Which statement is ALWAYS true?

- A. $P(A \cup B) = P(A) + P(B)$
- B. $P(A \cap B) \leq P(A)$
- C. $P(A) > P(A^c)$
- D. $P(A \cup B) \geq 1$

Q3. If $A \subset B$, then:

- A. $P(A) = P(B)$
- B. $P(A) \geq P(B)$
- C. $P(A) \leq P(B)$
- D. $P(A \cap B) = 0$

Q4. If $A \cap B = \emptyset$, then:

- A. Independent
- B. $P(A \cup B) = P(A) + P(B)$
- C. $P(A \cap B) = P(A)P(B)$
- D. $P(A) = P(B)$

Q5. Given $P(A)=0.4$, $P(B|A)=0.5$, $P(B)=0.3$, find $P(A|B)$.

- A. 0.66
- B. 0.5
- C. 0.4
- D. 0.25

Q6. If A and B are independent:

- A. $P(A|B) = P(B)$
- B. $P(A|B) = P(A)$
- C. $P(A \cap B) = 0$
- D. $P(A) = P(B)$

Q7. Disease prevalence 2%, sensitivity 0.9, specificity 0.95. Probability of positive test?

- A. 0.02
- B. 0.048
- C. 0.068
- D. 0.95

Q8. Using previous data, probability of disease given positive?

- A. ≈ 0.27
- B. ≈ 0.5
- C. ≈ 0.9
- D. ≈ 0.02

Q9. Probability of drawing a Queen or King from 52-card deck?

- A. $8/52$
- B. $4/52$
- C. $16/52$
- D. $12/52$

Q10. If $X \sim \text{Bin}(n, p)$, expected value is:

- A. np
- B. $np(1-p)$
- C. p
- D. n

Q11. In a group of 20 people, probability at least two share a birthday?

- A. ≈ 0.41
- B. ≈ 0.59
- C. ≈ 0.27
- D. ≈ 0.80

Q12. Number of ways to choose 3 from 10:

- A. 720
- B. 120
- C. 60
- D. 1000

Q13. Number of permutations of 5 objects:

- A. 25
- B. 120
- C. 60
- D. 10

Q14. Probability of exactly 2 heads in 4 tosses:

- A. $6/16$
- B. $4/16$
- C. $8/16$
- D. $2/16$

Q15. For standard normal, $P(Z < 0) =$

- A. 0
- B. 0.5
- C. 1
- D. 0.25

Q16. Z-score of $x=180$, mean=170, sd=5:

- A. 1
- B. 2
- C. 3
- D. 0.5

Q17. If $Z=2$, $P(Z > 2) \approx$

- A. 0.5
- B. 0.022
- C. 0.98
- D. 0.25

Q18. If $P(A)=1$:

- A. A impossible
- B. A certain
- C. $P(A^c)=1$
- D. $P(A)=0$

Q19. If $P(A)=0.8$, $P(B)=0.7$, minimum $P(A \cap B)$?

- A. 0
- B. 0.5
- C. 0.3
- D. 0.8

Q20. 6 people (3 boys, 3 girls). Probability boys sit together?

- A. $1/6$
- B. $1/5$
- C. $1/10$
- D. $1/20$

